

Adapting Schrödinger Bridge Flows for Constrained Compute: Curriculum-Enhanced α -DSBM

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Abstract

Finding a map to transport one probability distribution to another using unpaired data is a fundamental challenge in generative modeling. Recent advancements formulate this as a dynamic Schrödinger Bridge (SB) problem, yielding effective continuous-time diffusion models. The state-of-the-art approach, α -Diffusion Schrödinger Bridge Matching (α -DSBM) [De Bortoli et al., 2024], avoids the need to fully retrain multiple networks at each iteration, but its hardware requirements remain prohibitive, demanding large parameter counts and TPU clusters even for simple datasets like MNIST. This report investigates the behavior of α -DSBM under a fixed, severely constrained compute budget with a 5M-parameter model trained on limited data, and proposes **Curriculum-Enhanced α -DSBM** to address its failure modes. We show that under these constraints, the standard finetuning procedure collapses dramatically, with the model reverting to the initial distribution and failing to learn any meaningful transport. Our method, which introduces a *Leap Scheduler* for progressive target blending and an *Oscillating Scheduler* for directional loss reweighting, resolves this instability entirely. On 2D synthetic benchmarks where both methods converge, ours consistently achieves lower Wasserstein distance under the same compute budget, with gains of up to 35%. On constrained MNIST translation, it is the only method that remains stable. These results suggest that curriculum-based scheduling is a practical and principled path toward democratizing Entropic Optimal Transport techniques for resource-limited settings.

1 Introduction

The problem of finding a map to transport one probability distribution to another has numerous applications in machine learning. In particular, it is at the core of generative modeling—transforming a noise distribution into a data distribution—and is central to transfer learning tasks such as image-to-image translation. Generative Adversarial Networks (GANs) [Goodfellow et al., 2014] and Denoising Diffusion Models (DDMs) [Ho et al., 2020] have shown impressive performance in these domains.

When working with unpaired data samples from a source distribution π_0 and a target distribution π_1 , we aim to design a transport map from π_0 to π_1 . While standard Optimal Transport (OT) can be applied, it is computationally expensive. By showing that an entropy-regularised version of OT—the Entropic OT (EOT) [Cuturi, 2013]—could be computed much more efficiently, researchers have enabled transport ideas to be used in numerous applications.

To leverage the iterative refinement properties of DDMs in the OT setting, techniques to approximate the dynamic equivalent to EOT, the Schrödinger Bridge (SB), have been proposed [De Bortoli et al., 2021]. Among the most recent and promising is α -DSBM [De Bortoli et al., 2024]. However, scaling these continuous-time models requires prohibitive amounts of compute. In this report, we detail the mathematical foundations of α -DSBM, expose its hardware bottlenecks, and propose a novel curriculum-based scheduling algorithm that makes this framework viable for highly constrained hardware.

2 Background and Problem Formulation

2.1 Entropic Optimal Transport (EOT)

Given unpaired data samples, the transport problem corresponds to an unpaired data transfer task. We formulate this as finding a coupling distribution Π on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals $\Pi_0 = \pi_0$ and $\Pi_1 = \pi_1$. We seek the EOT coupling Π^* defined as:

$$\Pi^* = \operatorname{argmin}_{\Pi \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)} \left\{ \int_{\mathbb{R}^d \times \mathbb{R}^d} \frac{1}{2} \|x - y\|^2 d\Pi(x, y) - \epsilon H(\Pi) \right\}$$

where $H(\Pi)$ is the differential entropy of Π and $\epsilon > 0$ is a regularisation hyperparameter.

2.2 The Dynamic Schrödinger Bridge (SB)

To leverage recent advances in generative modeling, we turn to a dynamic formulation of EOT. The Schrödinger Bridge problem is defined as finding a continuous path measure $\mathbb{P}^* \in \mathcal{P}(\mathcal{C})$ such that:

$$\mathbb{P}^* = \operatorname{argmin}_{\mathbb{P} \in \mathcal{P}(\mathcal{C})} \{ \text{KL}(\mathbb{P} \parallel \mathbb{Q}) ; \mathbb{P}_0 = \pi_0, \mathbb{P}_1 = \pi_1 \}$$

with $\mathbb{Q} \in \mathcal{P}(\mathcal{C})$ induced by a scaled d -dimensional Brownian motion $(\sqrt{\epsilon} B_t)_{t \in [0,1]}$. Solving this dynamic problem is equivalent to optimizing the vector field of a stochastic process. The framework relies on two theoretical projections: the **Reciprocal Projection** ($\text{proj}_{\mathcal{R}}$), which ensures the conditional distribution of a path given its endpoints matches the reference process $\mathbb{Q}$; and the **Markovian Projection** ($\text{proj}_{\mathcal{M}}$), which preserves marginals and defines the drift field $v_t^*(x_t) = (\mathbb{E}_{\mathbb{P}}[X_1 | X_t = x_t] - x_t) / (1 - t)$.

3 The Original α -DSBM Framework

Older methods like Iterative Markovian Fitting (IMF) are expensive because they require training multiple DDM-type models. The α -DSBM algorithm [De Bortoli et al., 2024] improves upon this by taking a single gradient step to update the vector field, parameterized by $\alpha \in (0, 1]$. Furthermore, instead of using two separate networks for the forward and backward drifts, α -DSBM uses an additional input $s \in \{0, 1\}$ such that $v_{\theta}(1, \cdot)$ approximates the forward drift and $v_{\theta}(0, \cdot)$ approximates the backward drift, substantially reducing the number of parameters.

The training procedure consists of two stages: pretraining via bridge matching on independently drawn pairs, followed by online finetuning where the EMA network generates training samples on the fly. The interpolated bridge is given by $\hat{X}_t = (1 - t)X_0 + tX_1 + \sqrt{\epsilon t(1 - t)} Z$ with $Z \sim \mathcal{N}(0, I)$, and the model is trained to regress the drift targets $(\mathbf{X}_1 - \hat{\mathbf{X}}_t) / (1 - t)$ and $(\mathbf{X}_0 - \hat{\mathbf{X}}_t) / t$ in the forward and backward directions respectively. The full pseudocode is provided in Appendix A.

4 The Scaling Bottleneck and Hardware Constraints

While the theory behind α -DSBM is elegant, deploying it in practice reveals massive computational bottlenecks. For a seemingly simple image translation task (MNIST to EMNIST), the standard architecture utilizes a U-Net with **9 million parameters**. Pretraining takes **1 hour** on **4 v3 TPUs**, while the finetuning stage requires **4 hours** on **16 v3 TPUs**. Scaling to higher resolution datasets such as AFHQ-256 requires **226.7 million parameters**, with finetuning taking **37 hours** on 16 v3 TPUs.

These requirements make the standard algorithm impossible to run in our constrained, single-GPU environment. We drastically scaled down the network to $\sim 5\text{M}$ parameters with reduced train-

ing iterations. Under these constraints, the standard algorithm fails: the reduced capacity causes severe instability during finetuning, the network cannot simultaneously learn both drift directions accurately, leading to broken trajectories, mode collapse, and poor sample alignment. We attempted first universal regularizations, like using AdamW instead of Adam (used in the paper) and learning rate warmup, but still observed no stabilization.

5 Proposed Solution: Curriculum-Enhanced α -DSBM

To resolve the instability inherent in low-parameter environments, we developed **Curriculum-Enhanced α -DSBM**. We preserve the pretraining phase to establish a foundational vector field, but entirely redesign the finetuning phase using progressive scheduling. Inspired by curriculum strategies in reinforcement learning, where policies are trained on easier tasks before harder ones [Bengio et al., 2009], we introduce two dynamic scheduling variables.

5.1 Dynamic Scheduling Variables

Leap Scheduler (λ). Let n be the current step, N the total number of finetuning steps, $r \in (0, 1)$ a snap ratio, and $n_{\text{snap}} = \lfloor r \cdot N \rfloor$. The blending coefficient λ is defined as:

$$\lambda(n) = \begin{cases} \lambda_0 + (1 - \lambda_0) \cdot \frac{1 - \cos\left(\pi \cdot \min\left(1, \frac{n}{n_{\text{snap}}}\right)\right)}{2} & \text{if } n < n_{\text{snap}} \\ 1 & \text{if } n \geq n_{\text{snap}} \end{cases}$$

where λ_0 is the initial blending value (fixed at 0) and r is the snap ratio controlling what fraction of total steps the ramp covers before snapping to 1. In our experiments, the values reported as λ in the tables refer to this snap ratio r : e.g. $\lambda=0.25$ means the cosine ramp completes at 25% of total finetuning steps, after which full target blending is maintained. This allows the network to mostly target its own outputs early on, progressively “leaping” toward the true distributions as training stabilizes.

Oscillating Scheduler ($\alpha_{\text{osc}}, \beta_{\text{osc}}$). Let $n_{\text{snap}} = \lfloor r' \cdot N \rfloor$ with snap ratio r' , and define normalized time $\tau = n/n_{\text{snap}}$. The forward weight α_{osc} oscillates around 0.5 with exponentially decaying amplitude:

$$\alpha_{\text{osc}}(n) = \begin{cases} \text{clamp}(0.5 + 0.5 e^{-\delta\tau} \cdot \sin(2\pi f\tau + \phi), 0, 1) & \text{if } n < n_{\text{snap}} \\ 0.5 & \text{if } n \geq n_{\text{snap}} \end{cases}$$

$$\beta_{\text{osc}}(n) = 1 - \alpha_{\text{osc}}(n)$$

where f is the oscillation frequency, δ the decay rate, and $\phi \sim \text{Unif}([0, 2\pi))$ a random phase offset sampled at initialization. The exponential envelope $e^{-\delta\tau}$ ensures the oscillation dampens to zero, so both weights converge to 0.5 (equal weighting). This continuously shifts gradient emphasis between $\ell^{\rightarrow}$ and $\ell^{\leftarrow}$, preventing the low-capacity model from stagnating in either direction. What backed this intuition is that the original α -DSBM used two networks trained separately; the cosine schedule is a smoothed version of this that avoids catastrophic forgetting (both directions share the same parameters), while also ensuring $\alpha_{\text{osc}} + \beta_{\text{osc}} = 1$, which prevents any biased amplification of the gradients.

The full Curriculum-Enhanced finetuning procedure is given in Algorithm 1.

Algorithm 1 Curriculum-Enhanced α -DSBM (Finetuning Phase)

```
1: Input:  $\pi_0, \pi_1$ , Leap Sched. ( $\lambda$ ), Osc. Sched. ( $\alpha_{\text{osc}}, \beta_{\text{osc}}$ ),  $\gamma, \theta, \theta^{\text{EMA}}, \alpha$ 
2: [Pretraining phase identical to original  $\alpha$ -DSBM, see Appendix A]
3: for each finetuning step  $n$  do
4:   Sample  $(X_0, X_1) \sim (\pi_0 \otimes \pi_1)^{\otimes b}$ 
5:   % Simulate Noisy Targets (EMA)
6:    $\tilde{X}_1 \leftarrow$  Fwd SDE from  $X_0$  with  $v_{\theta^{\text{EMA}}}(1, \cdot)$ 
7:    $\tilde{X}_0 \leftarrow$  Bwd SDE from  $X_1$  with  $v_{\theta^{\text{EMA}}}(0, \cdot)$ 
8:   % Curriculum Variables
9:    $\lambda \leftarrow$  LeapScheduler.step()
10:   $\alpha_{\text{osc}}, \beta_{\text{osc}} \leftarrow$  OscSched.step()
11:  % Blend Distributions
12:   $X_{0,b} = (1-\lambda)X_0 + \lambda\tilde{X}_0$ ;  $X_{1,b} = (1-\lambda)X_1 + \lambda\tilde{X}_1$ 
13:   $X_{t,f} = \text{Interp}(X_{0,b}, X_{1,b}, Z^{\rightarrow})$ ;  $X_{t,b} = \text{Interp}(X_{0,b}, X_{1,b}, Z^{\leftarrow})$ 
14:  Compute  $\ell^{\rightarrow}$  and  $\ell^{\leftarrow}$  against target networks
15:  % Weighted Update
16:   $\mathcal{L} = \alpha_{\text{osc}} \ell^{\rightarrow} + \beta_{\text{osc}} \ell^{\leftarrow}$ 
17:   $\theta \leftarrow \theta - \alpha \nabla_{\theta} \mathcal{L}$ ;  $\theta^{\text{EMA}} \leftarrow \gamma \theta^{\text{EMA}} + (1-\gamma)\theta$ 
18: end for
```

6 Experiments and Observations

6.1 2D Toy Datasets Baseline

Before applying hardware constraints, we validated our codebase against standard 2D topological mapping tasks. In experiments translating “Blobs” to “Moons” and “S-Curve” to “Spiral”, the algorithm successfully recovered complex geometric structures. The generated samples matched the target distribution characteristics robustly, confirming that the foundational SDE solving and interpolation logic was sound. As shown in Figure 1, the model is not only able to perform the forward transport from Blobs to Moons, but the backward direction is equally accurate, which highlights a key strength of this method: it jointly learns both transport directions, a property that is essential in applications requiring invertible mappings.

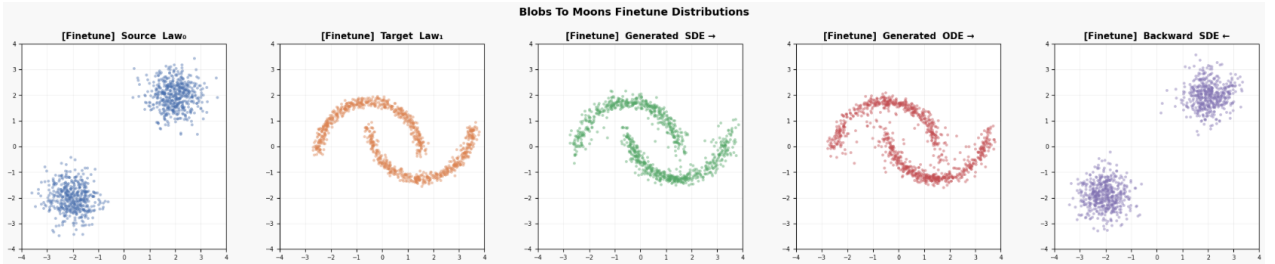


Figure 1: Blobs to Moons: forward and backward transport distributions after finetuning.

Table 1 reports W_2 and Path Energy (PE) across five 2D transport tasks (lower is better for both). The results consistently validate our curriculum approach: **every Lambda and Combined configuration outperforms the Base on W_2** across all five tasks, meaning that regardless of the specific hyperparameter choice, any instance of our method reliably improves over the standard finetuning baseline. The improvement is moreover monotonic in λ : a larger snap ratio yields progressively lower W_2 , meaning that completing the blending ramp over a greater fraction of total steps leads to more stable convergence, with $\lambda=0.25$ (ramp completing at 25% of steps) reducing the Wasserstein distance by up to 35% on Blobs $\rightarrow$ Moons (from .434 to .282). The Combined configurations are particularly noteworthy: by pairing both schedulers, they consistently match or exceed the best Lambda-only W_2 scores while simultaneously achieving lower Path Energy, confirming the

Table 1: W_2 ($\downarrow$) and Path Energy (PE, $\downarrow$) across all setups (20 ODE steps, 1000 samples, avg. 5 runs).

Method	<i>Blobs</i> → <i>Moons</i>		<i>Circles</i> → <i>Blobs</i>		<i>Swiss Rolls</i> → <i>Moons</i>		<i>S-Curve</i> → <i>Spirale</i>		<i>Moons</i> → <i>Circle</i>	
	W_2	PE	W_2	PE	W_2	PE	W_2	PE	W_2	PE
Pretrain	.836	5.901	.659	2.274	.594	1.165	.472	0.265	.722	1.053
<i>Base</i>										
Base	.434	5.762	.529	2.350	.310	1.312	.294	.712	.378	1.692
<i>Lambda</i>										
$\lambda=0.05$	.367	5.508	.480	2.193	.251	1.361	.283	.720	.338	1.562
$\lambda=0.15$	.326	5.578	.427	2.299	.236	1.422	.273	.712	.302	1.636
$\lambda=0.25$	.282	5.607	.436	2.281	.225	1.446	.265	.731	.274	1.731
<i>OSC</i>										
OSC $r=0.15$	.435	5.815	.656	2.416	.289	1.339	.293	.701	.381	1.724
OSC $r=0.50$	.434	5.731	.749	2.412	.291	1.323	.283	.715	.351	1.699
OSC $r=0.75$	.415	5.794	.693	2.414	.295	1.334	.320	.692	.355	1.718
<i>Combined</i>										
$\lambda=0.05, r=0.15$	.364	5.539	.468	2.344	.248	1.380	.289	.718	.336	1.576
$\lambda=0.15, r=0.50$	.328	5.546	.514	2.294	.241	1.397	.261	.722	.313	1.607
$\lambda=0.25, r=0.75$	.285	5.621	.408	2.213	.234	1.440	.265	.715	.277	1.726

two schedulers are complementary. The Oscillating Scheduler alone contributes modest W_2 gains but its primary benefit lies in producing more energy-efficient transport paths.

6.2 Constrained MNIST Translation

After successful 2D experiments, we transitioned to the MNIST image dataset using our restricted 5M-parameter architecture with a severely reduced number of finetuning iterations (batch size 128, 60k total training samples).

When running the **Standard α -DSBM** algorithm under these constraints, the model collapsed entirely. Specifically, the model reverted to outputting the initial samples X_0 , meaning the learned drift field converged to zero and no transport was performed. This failure mode persisted despite following the original algorithm’s guidelines exactly and attempting to replicate its architecture faithfully. Increasing the compute budget to 72 hours on a single P100 GPU did not resolve the collapse, confirming that the instability is structural rather than a consequence of insufficient training time.

In contrast, our **Curriculum-Enhanced** model maintained stability throughout finetuning. As reported in Table 2, both FID and MSD decrease steadily across finetuning checkpoints, demonstrating that the model continues to learn meaningful transport rather than degenerating. While the absolute FID values remain high given the severely constrained budget, the key result is the *absence of collapse*: the metrics improve monotonically, and as shown in Figure 2, the generated samples retain visible structure after 60 epochs of finetuning. Given sufficient compute, the observed trend strongly suggests convergence to a meaningful solution.

7 Conclusion

The α -DSBM framework [De Bortoli et al., 2024] offers a theoretically elegant, flow-based solution to Entropic Optimal Transport, building on foundational work on diffusion Schrödinger bridges

Table 2: FID ($\downarrow$) and MSD ($\downarrow$) on constrained MNIST translation across finetuning stages.

Stage	FID	MSD
Pretrained	196.39	0.2496
Finetuning epoch 30	170.98	0.1297
Finetuning Epoch 60	167.31	0.1469

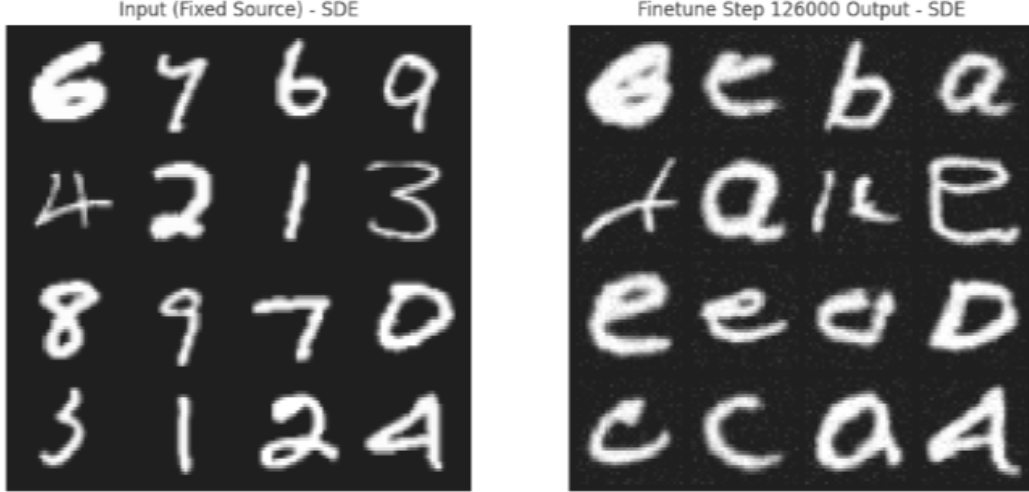


Figure 2: Generated MNIST samples after 60 epochs of Curriculum-Enhanced finetuning and 230 epochs of pretraining. Despite the constrained budget, the model maintains stable transport and produces structured outputs.

for unpaired data translation [De Bortoli et al., 2021] and on the broader generative modeling advances of GANs [Goodfellow et al., 2014] and DDMs [Ho et al., 2020]. However, its native design relies heavily on over-parameterization and immense compute to maintain stability during online finetuning. By diagnosing this hardware bottleneck and introducing a Curriculum-Enhanced finetuning strategy, our dynamic scheduling mechanism—a Leap Scheduler for target blending and an Oscillating Scheduler for directional loss weighting—resolves the catastrophic collapse of standard finetuning and consistently outperforms the baseline on 2D benchmarks, with Wasserstein gains of up to 35%. These results suggest that curriculum-based scheduling is a practical and principled path toward democratizing Schrödinger Bridge methods for resource-limited settings.

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A Original α -DSBM Algorithm

For completeness, we reproduce the full training procedure of the original α -DSBM [De Bortoli et al., 2024] below, including the explicit training loss definitions. Our Curriculum-Enhanced method (Algorithm 1 in the main text) replaces only the finetuning loop; the pretraining phase is identical.

The empirical training objectives are:

$$\begin{aligned}\ell^{\rightarrow}(\theta; t, \mathbf{X}_1, \hat{\mathbf{X}}_t) &= \frac{1}{B} \sum_{i=1}^B \left\| v_{\theta}^{\rightarrow}(t^i, \hat{\mathbf{X}}_t^i) - \frac{\mathbf{X}_1^i - \hat{\mathbf{X}}_t^i}{1 - t^i} \right\|^2 \\ \ell^{\leftarrow}(\theta; t, \mathbf{X}_0, \hat{\mathbf{X}}_t) &= \frac{1}{B} \sum_{i=1}^B \left\| v_{\theta}^{\leftarrow}(1 - t^i, \hat{\mathbf{X}}_t^i) - \frac{\mathbf{X}_0^i - \hat{\mathbf{X}}_t^i}{t^i} \right\|^2\end{aligned}$$

Algorithm 2 Original α -Diffusion Schrödinger Bridge Matching [De Bortoli et al., 2024]

- 1: **Input:** datasets π_0 and π_1 , entropic regularisation ϵ , steps N_{pre} and N_{fine} , batch size B , half batch size $b = B/2$, EMA decay γ , parameters θ , $\theta^{\text{EMA}} = \theta$, $\alpha \in (0, 1]$
 - 2: **for** $n \in \{1, \dots, N_{\text{pre}}\}$ **do**
 - 3: Sample $(\mathbf{X}_0, \mathbf{X}_1) \sim (\pi_0 \otimes \pi_1)^{\otimes B}$
 - 4: Sample $t \sim \text{Unif}([0, 1])^{\otimes B}$, $\mathbf{Z} \sim \mathcal{N}(0, \text{Id})^{\otimes B}$; compute $\mathbf{X}_t = \text{Interp}_t(\mathbf{X}_0, \mathbf{X}_1, \mathbf{Z})$
 - 5: Update θ with gradient step on $\frac{1}{2} [\ell^{\rightarrow}(t^{1:b}, \mathbf{X}_1^{1:b}, \mathbf{X}_t^{1:b}) + \ell^{\leftarrow}(t^{b+1:B}, \mathbf{X}_0^{b+1:B}, \mathbf{X}_t^{b+1:B})]$
 - 6: Update EMA: $\theta^{\text{EMA}} = \gamma \theta^{\text{EMA}} + (1 - \gamma) \theta$
 - 7: **end for**
 - 8: **for** $n \in \{1, \dots, N_{\text{fine}}\}$ **do**
 - 9: Sample $(\mathbf{X}_0, \mathbf{X}_1) \sim (\pi_0 \otimes \pi_1)^{\otimes b}$
 - 10: Sample $\hat{\mathbf{X}}_1$ from forward SDE with $v_{\theta^{\text{EMA}}}(1, \cdot)$ starting from $\mathbf{X}_0$
 - 11: Sample $\hat{\mathbf{X}}_0$ from backward SDE with $v_{\theta^{\text{EMA}}}(0, \cdot)$ starting from $\mathbf{X}_1$
 - 12: Compute $\mathbf{X}_t^{\rightarrow} = \text{Interp}_{t^{\rightarrow}}(\hat{\mathbf{X}}_0, \mathbf{X}_1, \mathbf{Z}^{\rightarrow})$ and $\mathbf{X}_t^{\leftarrow} = \text{Interp}_{t^{\leftarrow}}(\mathbf{X}_0, \hat{\mathbf{X}}_1, \mathbf{Z}^{\leftarrow})$
 - 13: Update θ with gradient step on $\frac{1}{2} [\ell^{\rightarrow}(t^{\rightarrow}, \mathbf{X}_1, \mathbf{X}_t^{\rightarrow}) + \ell^{\leftarrow}(t^{\leftarrow}, \mathbf{X}_0, \mathbf{X}_t^{\leftarrow})]$ with stepsize α
 - 14: Update EMA: $\theta^{\text{EMA}} = \gamma \theta^{\text{EMA}} + (1 - \gamma) \theta$
 - 15: **end for**
 - 16: **Output:** $(\theta, \theta^{\text{EMA}})$
-